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Course/Section: CPE31S1	Date Submitted: 19/02/2026
Instructor: Engr. Robin Valenzuela	Semester and SY: 2nd, 2025-2026

Activity 3: Install SSH server on CentOS or RHEL 8

1. **Objectives:**

1.1 Install Community Enterprise OS or Red Hat Linux OS

1.2 Configure remote SSH connection from remote computer to CentOS/RHEL-8

2. **Discussion:**

CentOS vs. Debian: Overview

CentOS and Debian are Linux distributions that spawn from opposite ends of the candle.

CentOS is a free downstream rebuild of the commercial Red Hat Enterprise Linux distribution where, in contrast, Debian is the free upstream distribution that is the base for other distributions, including the Ubuntu Linux distribution.

As with many Linux distributions, CentOS and Debian are generally more alike than different; it isn't until we dig a little deeper that we find where they branch.

CentOS vs. Debian: Architecture

The available supported architectures can be the determining factor as to whether a distro is a viable option or not. Debian and CentOS are both very popular for x86_64/AMD64, but what other archs are supported by each?

Both Debian and CentOS support AArch64/ARM64, armhf/armhfp , i386 , ppc64el/ppc64le. (Note: armhf/armhfp and i386 are supported in CentOS 7 only.)

CentOS 7 additionally supports POWER9 while Debian and CentOS 8 do not. CentOS 7 focuses on the x86_64/AMD64 architecture with the other archs released through the AltArch SIG (Alternate Architecture Special Interest Group) with CentOS 8 supporting x86_64/AMD64, AArch64 and ppc64le equally.

Debian supports MIPSel, MIPS64el and s390x while CentOS does not. Much like CentOS 8, Debian does not favor one arch over another —all supported architectures are supported equally.

CentOS vs. Debian: Package Management

Most Linux distributions have some form of package manager nowadays, with some more complex and feature-rich than others.

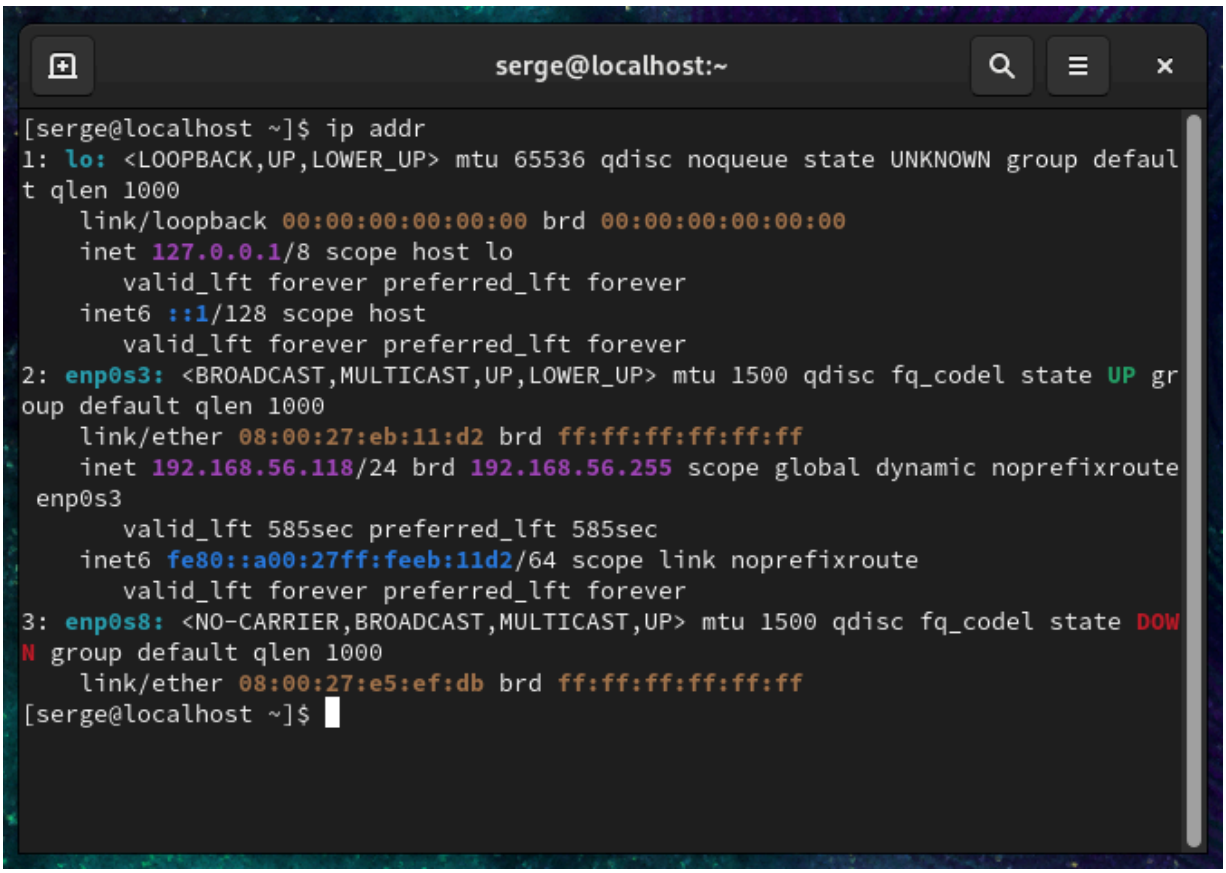
CentOS uses the RPM package format and YUM/DNF as the package manager.

Debian uses the DEB package format and dpkg/APT as the package manager.

Both offer full-feature package management with network-based repository support, dependency checking and resolution, etc.. If you're familiar with one but not the other, you may have a little trouble switching over, but they're not overwhelmingly different. They both have similar features, just available through a different interface.

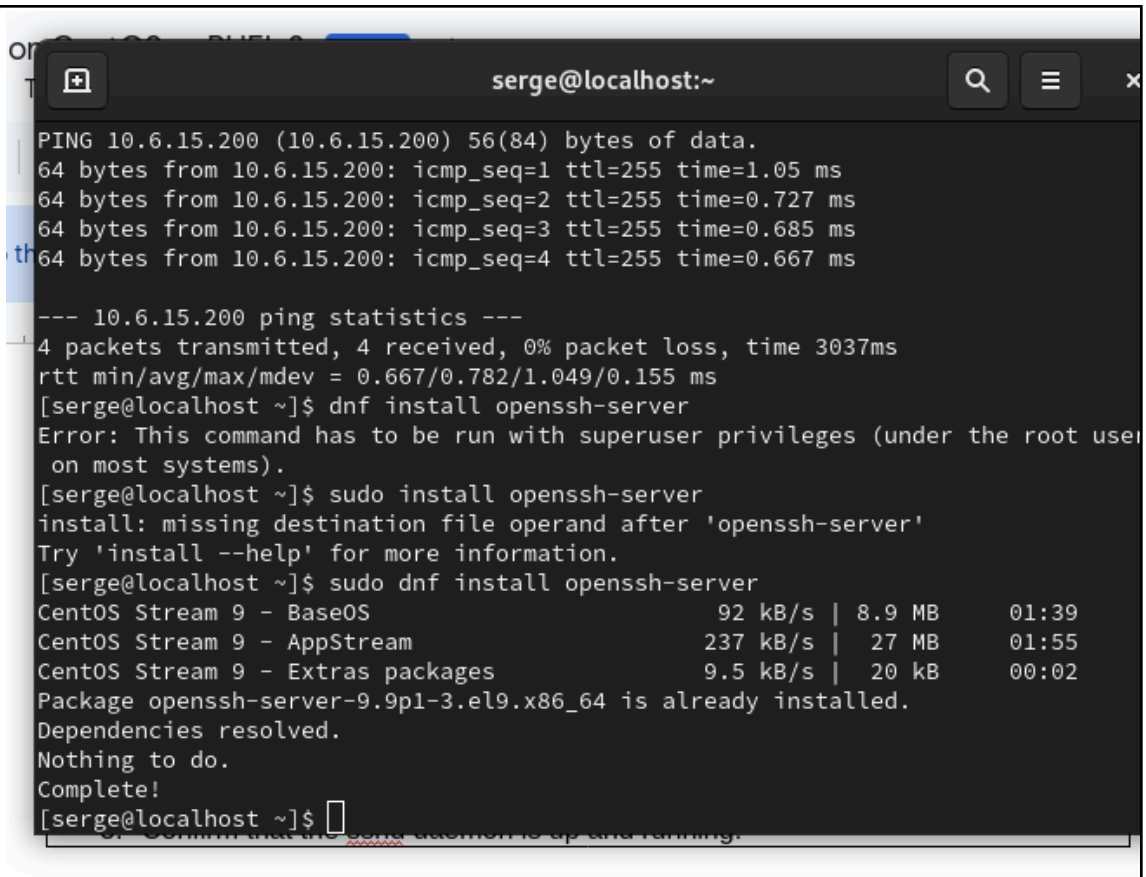
Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here: http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.



Task 2: Install the SSH server package *openssh*

1. Install the ssh server package *openssh* by using the *dnf* command:
   $ dnf install openssh-server



```
serge@localhost:~
PING 10.6.15.200 (10.6.15.200) 56(84) bytes of data.
64 bytes from 10.6.15.200: icmp_seq=1 ttl=255 time=1.05 ms
64 bytes from 10.6.15.200: icmp_seq=2 ttl=255 time=0.727 ms
64 bytes from 10.6.15.200: icmp_seq=3 ttl=255 time=0.685 ms
64 bytes from 10.6.15.200: icmp_seq=4 ttl=255 time=0.667 ms

--- 10.6.15.200 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3037ms
rtt min/avg/max/mdev = 0.667/0.782/1.049/0.155 ms
[serge@localhost ~]$ dnf install openssh-server
Error: This command has to be run with superuser privileges (under the root user
 on most systems).
[serge@localhost ~]$ sudo install openssh-server
install: missing destination file operand after 'openssh-server'
Try 'install --help' for more information.
[serge@localhost ~]$ sudo dnf install openssh-server
CentOS Stream 9 - BaseOS                         92 kB/s | 8.9 MB     01:39
CentOS Stream 9 - AppStream                     237 kB/s |  27 MB     01:55
CentOS Stream 9 - Extras packages               9.5 kB/s |  20 kB     00:02
Package openssh-server-9.9p1-3.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[serge@localhost ~]$
```

2. Start the *sshd* daemon and set to start after reboot:
   $ systemctl start sshd
   $ systemctl enable sshd
3. Confirm that the sshd daemon is up and running:
   $ systemctl status sshd

```
serge@localhost:~
[serge@localhost ~]$ systemctl start sshd
[serge@localhost ~]$ sudo systemctl enable sshd
[sudo] password for serge:
[serge@localhost ~]$ sudo systemctl status sshd
Failed to find catalog entry: Invalid argument
Failed to find catalog entry: Invalid argument
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena
     Active: active (running) since Thu 2026-02-19 17:45:44 PST; 16min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 974 (sshd)
      Tasks: 1 (limit: 22965)
     Memory: 2.5M (peak: 2.7M)
        CPU: 10ms
     CGroup: /system.slice/sshd.service
             └─974 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 19 17:45:44 localhost.localdomain systemd[1]: Starting OpenSSH server daemo
Feb 19 17:45:44 localhost.localdomain sshd[974]: Server listening on 0.0.0.0 po
Feb 19 17:45:44 localhost.localdomain systemd[1]: Started OpenSSH server daemon
Feb 19 17:45:44 localhost.localdomain sshd[974]: Server listening on :: port 22

[serge@localhost ~]$
```

4. Open the SSH port 22 to allow incoming traffic:
   $ firewall-cmd --zone=public --permanent --add-service=ssh
   $ firewall-cmd --reload

```
serge@localhost:~
Failed to find catalog entry: Invalid argument
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena
     Active: active (running) since Thu 2026-02-19 17:45:44 PST; 16min ago
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Feb 19 17:45:44 localhost.localdomain sshd[974]: Server listening on :: port 22

[serge@localhost ~]$ firewall-cmd --zone=public --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[serge@localhost ~]$ firewall-cmd --reload
success
[serge@localhost ~]$
```
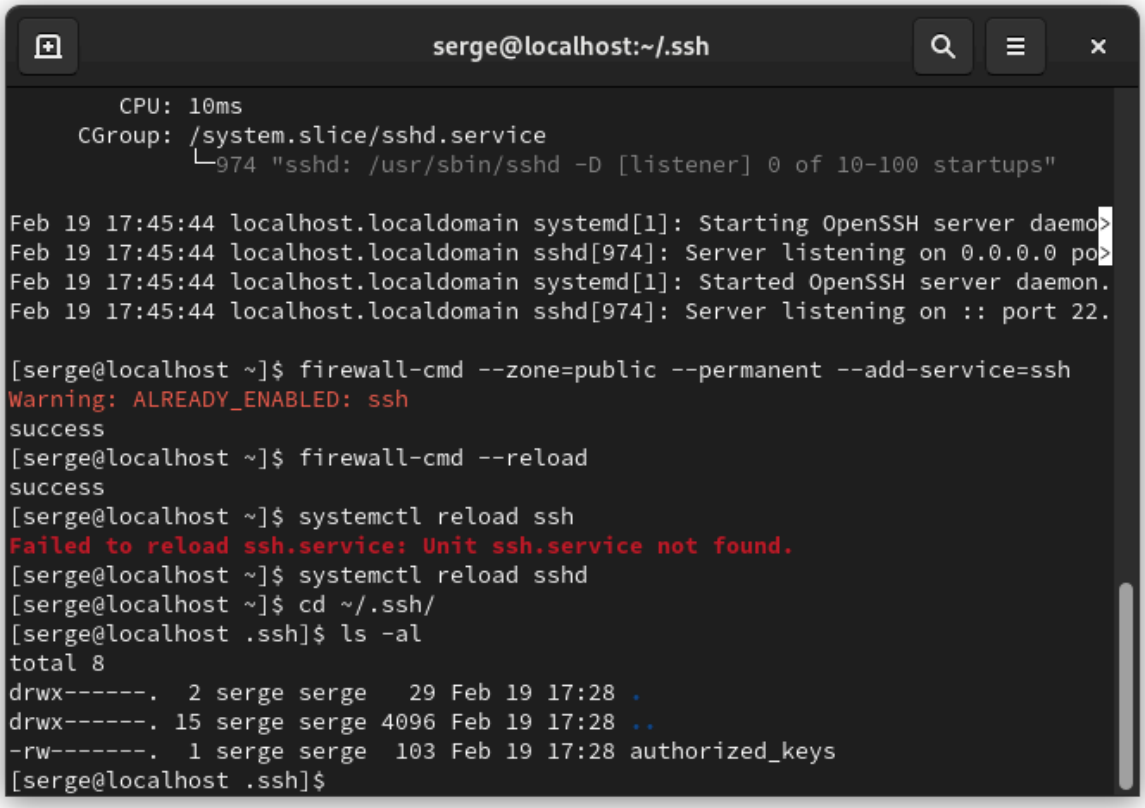

5. Locate the ssh server man config file /*etc/ssh/sshd_config* and perform custom configuration. Every time you make any change to the /*etc/ssh/sshd-config* configuration file reload the *sshd* service to apply changes:
   $ systemctl reload sshd

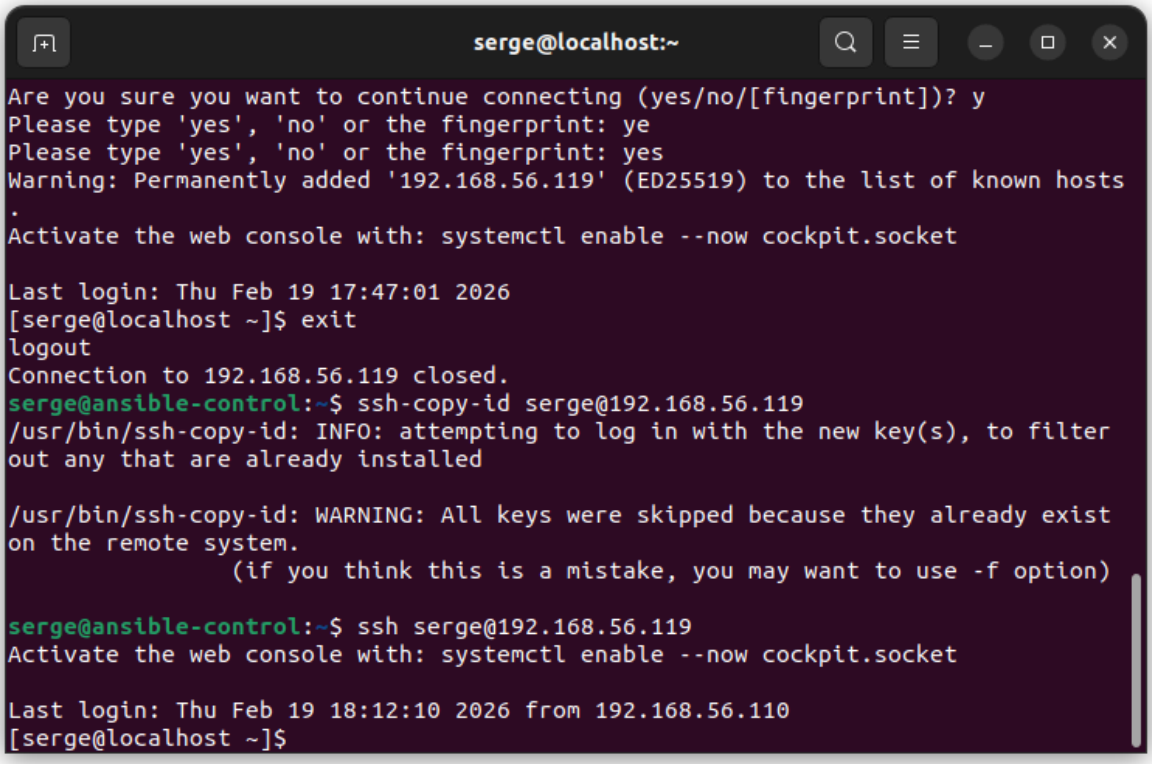
Task 3: Copy the Public Key to CentOS

1. Make sure that *ssh* is installed on the local machine.
2. Using the command *ssh-copy-id*, connect your local machine to CentOS.
3. On CentOS, verify that you have the *authorized_keys*.



Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected.



Reflections:

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

I'd look at the availability of packages and software, and how easy it is to manage them. I'd also look at their stability and security.

2. What are the main diffence between Debian and Red Hat Linux distributions?

They mainly differ in terms of stability, community, package manageability, and release cycles/life cycles.The system administration tools and desktop environments also have their differences.